

Valuing prevention: a synthetic microsimulation of the fiscal savings from Social Sundhed’s *Social Brobygger* initiative among young disability-benefit recipients (18–30)

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Abstract

This project builds a transparent, fully synthetic microsimulation that estimates the present value of public expenditure that could be *avoided* if Social Sundhed’s volunteer “Social Brobygger” (social bridge-building) initiative raises the probability that young Danes (18–30) on *førtidspension* (FP, disability pension) or in a *ressourceforløb* (RES, time-limited resource-clarification course) move into work, fleksjob or off benefits. The framing deliberately mirrors the state’s valuation of the mink-compensation case (29.2 DKK bn, FVM Jan-2025), where the present value of long-horizon future streams was the object of interest. Using literature- and register-anchored parameters and two layers of uncertainty analysis (a one-way tornado and a full probabilistic sensitivity analysis), the model produces a mean saving of roughly 32,000 DKK per participant, concentrated almost entirely in the *ressourceforløb* track. Under joint parameter uncertainty the saving is positive in about 94% of draws, but its *magnitude* is driven mainly by the fleksjob-versus-ordinary-job destination mix and the relapse rate rather than by the treatment effect alone. All data are synthetic; every parameter is documented and reproducible.

Contents

1 Research question and motivation

Denmark re-evaluates many awarded disability pensions every three years, and since the 2013 reform most under-40s are routed through a *ressourceforløb* rather than granted a permanent pension outright. Both facts open a policy question with a clear fiscal dimension:

How much future public expenditure could a low-cost, volunteer-driven bridging initiative plausibly avoid among 18–30-year-olds on FP or in a ressourceforløb, if it raises their chance of moving into work or fleksjob (and, for RES, lowers their chance of progressing into permanent FP)?

The single most expensive event for this age group is *progression into permanent FP*: a 25-year-old who lands on FP can draw benefits for roughly 45 years. Preventing or reversing that draws a very long, discounted expenditure stream off the public books — which is exactly what we value.

The mink analogy. When the Danish state compensated mink farmers (29.2 DKK bn, Ministry of Food, Agriculture and Fisheries, Jan-2025), it valued the *present value of lost future income streams*. We use the mirror image: the present value of *avoided future expenditure streams*. Because the cohort is young, the horizon to folkepension age is 40–52 years, so — just as in the mink case — the result is dominated by the discount schedule. We therefore make discounting explicit and stress-test it.

2 Data: a synthetic register

Because individual-level social-register data are confidential, we *generate* a synthetic person-level dataset that imitates the structure of a Danish administrative extract: one person-row, typed variables, and a synthetic pseudo-identifier. It contains no real personal data. The generator (`export_synthetic_register.py`) writes `data/synthetic_cohort.csv` together with a machine-readable codebook (`data/codebook.csv`); the human-readable data dictionary is in `docs/data_dictionary.md`. Table ?? lists the core variables.

Table 1: Core variables in the synthetic cohort (one row per individual).

Variable	Type	Meaning
<code>age</code>	int	Age 18–30 at simulation start
<code>civilstatus</code>	cat.	<code>enlig</code> / <code>samlevende</code> (drives benefit rate)
<code>track</code>	cat.	<code>foertidspension</code> / <code>ressourceforloeb</code>
<code>benefit_gross_dkk</code>	int	Annual gross transfer in the start state
<code>net_annual_cost_dkk</code>	int	Gross $\times (1 - \text{tax_back})$
<code>fp_responsive</code>	0/1	Whether FP work capacity can develop
<code>res_years_left</code>	int	Remaining <code>ressourceforloeb</code> duration (0 if FP)
<code>years_to_folkepension</code>	int	Model horizon T_i
<code>npv_saving_dkk</code>	int	Monte-Carlo mean NPV saving, net of programme cost

3 Model

Each synthetic individual is simulated year-by-year from their start age to folkepension age as a Markov process over five states:

FP, RES, EMP (employment), FLEX (fleksjob), OTH (other benefit).

Annual transitions.

- RES $\rightarrow$ work/fleksjob with probability $q_{\text{res,work}}$ (an exit, split across EMP/FLEX/OTH);
- RES $\rightarrow$ FP progression with probability $q_{\text{res,fp}}$ (the expensive event);
- on RES expiry (course runs out unresolved) $\rightarrow$ FP with probability `res_expiry_to_fp`, else an exit;
- FP $\rightarrow$ exit *only* at the mandatory three-yearly review, with probability q_{fp} , and only for the *responsive* subgroup;
- relapse EMP/FLEX/OTH $\rightarrow$ FP with annual probability `relapse_per_year`.

Two arms. We run the identical cohort under a *control* arm (no initiative) and a *treated* arm (initiative), which differ only in the three policy-sensitive probabilities, summarised in Table ??.

Table 2: The two simulated arms differ only in three probabilities.

Arm	q_{fp} (per review)	$q_{res,work}$ (annual)	$q_{res,fp}$ (annual)
Control (no initiative)	0.015	0.10	0.10
Treated (initiative)	0.030	0.15	0.075

Fiscal accounting. Each state carries a net annual public cost (benefit minus tax clawed back; ordinary employment is *negative* because the state receives net tax). The saving for individual i is the difference in discounted lifetime public cost between the arms, net of the programme’s own cost C_{prog} :

$$\text{Saving}_i = \underbrace{PV_i^{\text{control}} - PV_i^{\text{treated}}}_{\text{avoided net expenditure}} - C_{\text{prog}}, \quad PV_i = \sum_{t=0}^{T_i} \text{cost}_{i,t} \cdot DF_t, \quad (1)$$

with cumulative discount factor $DF_t = \prod_{s=1}^t (1 + r_s)^{-1}$. The marginal rate r_s follows Finansministeriet’s declining schedule (3.5 % for years 1–35, 2.5 % for years 36–70, 1.5 % thereafter). Figure ?? shows how aggressively this erodes far-future kroner, which is why the horizon assumptions matter.

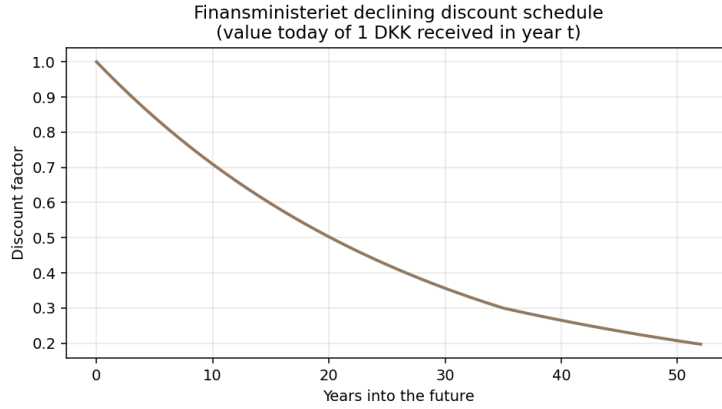


Figure 1: Finansministeriet declining discount schedule: present value today of one krone received in year t .

4 Evidence anchoring of the parameters

The transition and destination parameters are *directional anchors* drawn from named public sources, not exact register extractions for the 18–30 cohort. They should be replaced with precise figures before any external claim; the point of the model is to make each assumption explicit and testable.

Table 3: Literature/register-anchored default parameters.

Parameter	Default	Anchor
share_permanent	0.55	(S1) Post-2013 young FP grants mostly permanent; revocations rare
q_{fp} control / treated	0.015 / 0.030	(S1, S3) Revocation rare; initiative $\approx$ doubles it
dest. split (emp/fleks/other)	0.20 / 0.45 / 0.35	(S2) fleksjob dominates, ordinary jobs a minority
$q_{res,work}$ control / treated	0.10 / 0.15	(S2, S3) $\sim \frac{1}{4}$ of completed forløb reach work/fleksjob
$q_{res,fp}$ control / treated	0.10 / 0.075	(S2) $\sim \frac{1}{3}$ progress to FP; initiative prevents some

Sources.

(S1)

Reform of førtidspension and fleksjob (Lov nr. 1380 af 23/12/2012, in force 1 Jan 2013) and DST/STAR statistics on revocations (frakendelser).

(S2)

STAR / jobindsats.dk result statistics for completed ressourceforløb; Rigsrevisionen, *Beretning om ressourceforløb* (9/2019).

(S3)

IPS (Individual Placement and Support) evidence — international meta-analyses and Danish IPS trials — employment effects of about +10 to +15 percentage points, used as an *upper* anchor for the initiative’s effect. A volunteer bridging model is lighter-touch than full IPS, so defaults sit in the lower part of that range.

5 Results

5.1 Headline

With the anchored defaults the model yields the figures in Table ???. The per-participant saving is modest and highly skewed: most individuals never exit (so their saving is just the negative programme cost), while a minority of successful exits carry large, long-horizon savings. Crucially, the value is concentrated in the ressourceforløb track, consistent with the intuition that preventing *progression into* permanent FP is worth far more than reversing an already-granted FP.

Table 4: Headline results, anchored defaults (synthetic cohort, $n = 1,500$).

Quantity	Value
Mean NPV saving / participant	32,000 DKK (approx.)
Median NPV saving / participant	10,000 DKK (approx.)
Share of Monte-Carlo draws net-positive	27%
Mean saving — førtidspension track	11,000 DKK (approx.)
Mean saving — ressourceforløb track	52,000 DKK (approx.)

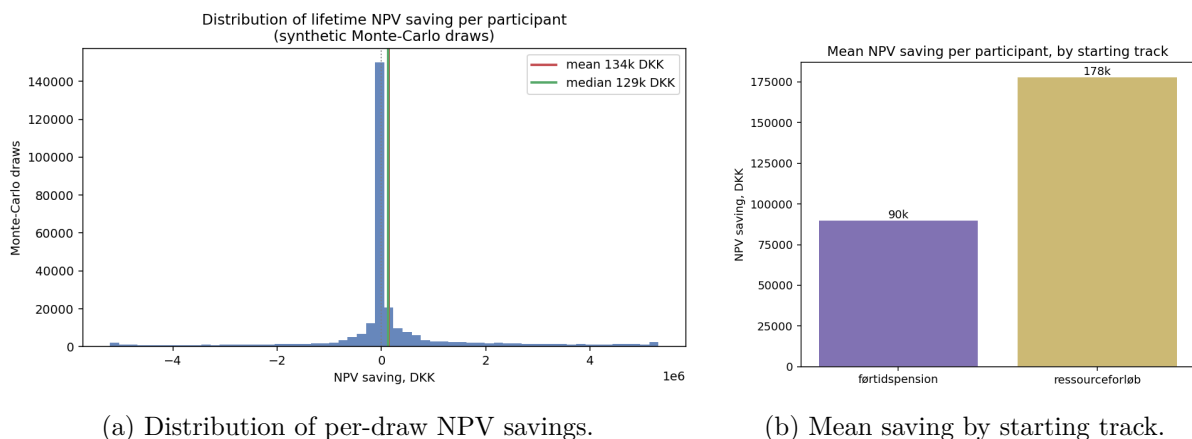


Figure 2: The saving is right-skewed (a) and concentrated in the ressourceforløb track (b).

5.2 One-way sensitivity (tornado)

Varying one assumption at a time (Figure ??) shows the result is most sensitive to the FP exit treatment effect Δq and the discount rate, then the RES→work effect and the permanent share.

This ranking is more defensible than the headline level itself.

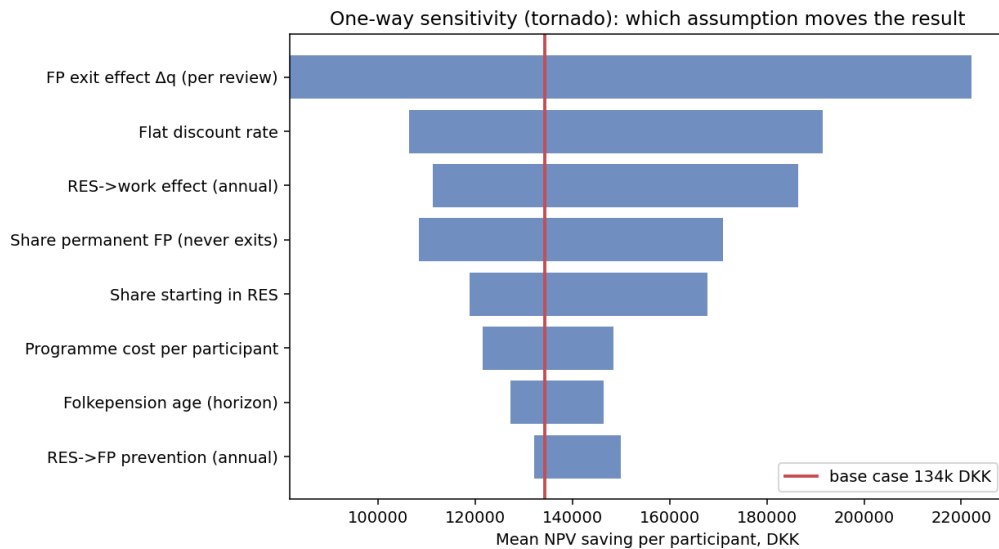


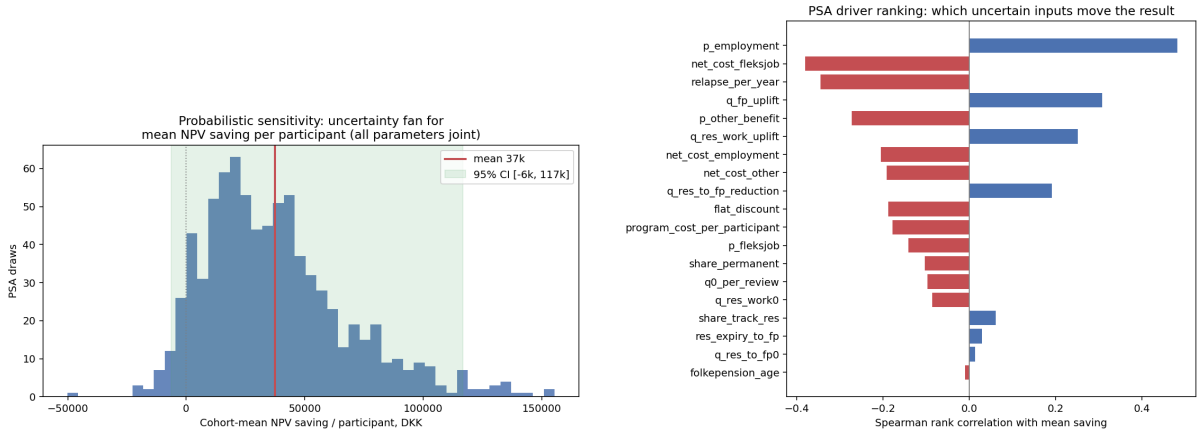
Figure 3: One-way sensitivity. Bars span the low/high value of the mean saving as each driver is varied alone; the red line is the base case.

5.3 Probabilistic sensitivity analysis

The PSA (`probabilistic_sensitivity.py`) samples *all* uncertain parameters jointly — Beta distributions for probabilities, a Dirichlet for the destination split, Normal for costs, and Uniform for the treatment uplifts and discount rate — and reruns the simulation 800 times. The resulting uncertainty fan (Figure ??) and driver ranking (Figure ??) give the report’s most honest summary:

- mean of the mean-saving $\approx 37,000$ DKK;
- 95% credible interval $\approx [-6,000 \text{ DKK}; 117,000 \text{ DKK}]$;
- $P(\text{saving} > 0) \approx 94\%$.

So even under joint uncertainty the initiative is very likely to save money, but the *size* of the saving is governed mainly by the fleksjob-versus-ordinary-employment destination mix (`p_employment`, `net_cost_fleksjob`) and the relapse rate — not by the treatment effect alone. That is where better Danish register data would most improve the estimate.



(a) Uncertainty fan for the mean saving.

(b) Driver ranking (rank correlation).

Figure 4: Probabilistic sensitivity analysis: the full joint-uncertainty fan (a) and which inputs move the result (b).

6 Limitations

1. **Synthetic data.** Levels are illustrative; the anchored parameters are directional, not exact register extractions for the 18–30 cohort.
2. **No causal identification.** The treatment effect Δq is an input anchored to IPS evidence, not an estimate for *this* programme. A credible own estimate would need an RCT or quasi-experiment.
3. **Selection.** Participants of a voluntary programme differ from non-participants; a naive Δq overstates savings.
4. **Partial fiscal scope.** We count benefits and income tax only — not health-care use, VAT, or quality-of-life gains, which would *raise* the social value.
5. **Horizon risk.** Folkepension age and the discount schedule are indexed and uncertain; both are stress-tested in the tornado and the PSA.

7 Reproducibility

All randomness is controlled by a fixed seed. The full pipeline is:

```
python early_retirement_savings_model.py # headline + tornado table
python export_synthetic_register.py      # data/synthetic_cohort.csv + codebook
python make_figures.py                  # figures/*.png
python probabilistic_sensitivity.py      # PSA fan, driver ranking, psa_draws.csv
```

Conclusion. Framed as a present-value calculation in the same spirit as the mink compensation, a low-cost bridging initiative for young disability-benefit recipients is very likely to be fiscally self-financing, with the bulk of the value coming from *preventing progression into permanent førtidspension* via the ressourceforløb track. The size of the prize, however, hinges on where successful participants land — a question only Danish register data can settle.